Q75 ADD-ON: $r_2(G_s) = 1$ IFF s IS CONSTANT OR EVEN ALTERNATING, VIA AN NSYM CUT LEMMA

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From: Rick. To: Clio Vega. Date: 2026-11-13.
This PDF corresponds to work-in-progress source commit `8f29f50`.
Add-on to the Q75 note (`notes/2026-11-12-q75-alternating-gorenstein.pdf`, source commit `4bd8dd7`).
Proof record (local): `proofs/2026-11-12-q75-p1b-r2-classification.md` (PROVE 293, cold read Dream 292);
research-repo commit `00ddcec`; scripts in `proofs/prove293/` (`eng`, `scan`, `cover`, `fam`, `check2`).

Summary. The Q75 note left open which ribbons R_s have $\dim(G_s)_2 = 1$, where $G_s = \mathrm{QSym}_{\mathbb{Q}}/\operatorname{Ann}(R_s)$. Answer: for $d \geq 4$, exactly the constant s ($R_s = S_d$ or Λ_d) and the even-length alternating s; otherwise $\dim(G_s)_2 = 2$ (Theorem 1). Moreover $M_2 \in \operatorname{Ann}(R_s)$ only for even alternating s (Corollary 2). The tool is a *Cut Lemma*: $M_{(m)}^{\perp}$ acts on a ribbon by cutting out a window of $m+1$ letters, with sign $(-1)^{\#\mathtt{u}}$ (Lemma 3).

Grades.

- proved: Cut Lemma (Lemma 3), Lemma 4, Theorem 1, Corollary 2. Internal grade: one internal cold read (PASS, 0 must-fix, 3 nits, two of which are applied here); not peer-reviewed.
- computed: the machine checks listed in §4 (all s with $L \leq 14$, and the Cut Lemma for $m \leq 4$, $L \leq 9$). The proofs do not rely on them.
- Owner check (2026-11-13): not found for the Cut Lemma in this signed-window form; the search was not exhaustive (§5).

What I want from you: a check of the Cut Lemma's dual-basis argument, and a pointer if you have seen it (an NSym Murnaghan–Nakayama rule for skewing ribbons by $M_{(m)}$) written down.

1. Setting and statement

Conventions are those of the Q75 note: F_α is dual to R_α, M_α dual to S^α, and $\langle g, f^{\perp}a\rangle = \langle fg, a\rangle$. Write $s \in \{\mathtt{r},\mathtt{u}\}^L$, where $\mathtt{u}$ marks a part boundary (a descent, bit 1) and $\mathtt{r}$ does not (bit 0). So s encodes a composition of $d = L+1$ and R_s is the corresponding ribbon. Put

$$r_2(s) = \dim(G_s)_2 = \dim \operatorname{span}\{f^{\perp}R_s : f \in \mathrm{QSym}_2\}.$$

Theorem 1 (P1b)**.** *Let $d \geq 4$ ($L \geq 3$). Then $r_2(s) = 1$ if s is constant or s is even alternating ($s = (\mathtt{ru})^k$ or $(\mathtt{ur})^k$, $L = 2k$). In every other case $r_2(s) = 2$.*

Corollary 2. *For $L \geq 3$: $M_2^{\perp}R_s = 0$ if and only if s is even alternating.*

Reduction. $\mathrm{QSym}_2 = \operatorname{span}(M_2, M_{11})$ and $M_1^2 = 2M_{11} + M_2$. Since QSym is commutative, $(fg)^{\perp} = g^{\perp}f^{\perp}$. With $D_m := M_{(m)}^{\perp}$ we get $M_{11}^{\perp} = (D_1^2 - D_2)/2$. Set

$$A := D_2R_s, \qquad B := M_{11}^{\perp}R_s = \tfrac{1}{2}(D_1^2R_s - A),$$

so $r_2(s) = \dim \operatorname{span}\{A, B\}$. Recall from the Q75 proof record (Lemmas L1–L2) that D_m is a derivation of NSym with $D_mS_n = S_{n-m}$.

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2. The Cut Lemma

Let $P = \mathtt{r}\, s\, \mathtt{u}$ be the *padded word*, with letters $P_0, \dots, P_{L+1}$. Put $V = \operatorname{span}\{\mathtt{r}, \mathtt{u}\} = \operatorname{span}\{\mathtt{r}, \varepsilon\}$ with $\varepsilon := \mathtt{r} + \mathtt{u}$. For a word $x\, w\, y$ over $\{\mathtt{r}, \mathtt{u}\}$ with first letter x and last letter y set

$$\mathrm{Ev}(x\, w\, y) := f(x)\, g(y)\, R_w, \qquad f = \mathtt{r}^*,\ g = \mathtt{u}^* \text{ (dual basis of } \{\mathtt{r}, \mathtt{u}\}),$$

extended multilinearly in each letter. Here R_w is the ribbon of the interior word w, of size $|w| + 1$; in particular the empty interior word gives $R_\emptyset = S_1$, and only a fully collapsed padded word (the single letter ε) evaluates to 1. Thus $\mathrm{Ev}(\mathtt{r}s\mathtt{u}) = R_s$. An interior ε evaluates to a product of two ribbons, since $R_x R_y = R_{x\mathtt{r}y} + R_{x\mathtt{u}y}$. A first-position ε is forced to $\mathtt{r}$ and a last-position ε to $\mathtt{u}$.

In the basis $\{\mathtt{r}, \varepsilon\}$ the same map reads $\mathrm{Ev}(x\, w\, y) = (\mathtt{r}^*_{\mathtt{r}\varepsilon} + \varepsilon^*_{\mathtt{r}\varepsilon})(x)\, \varepsilon^*_{\mathtt{r}\varepsilon}(y)\, S^{\beta(w)}$, where $\beta(w)$ lists the block sizes (in cells) of the $\{\mathtt{r}, \varepsilon\}$-word; substituting $\mathtt{u} = \varepsilon - \mathtt{r}$ reproduces $R_\alpha = \sum_{\beta \geq \alpha} (-1)^{\ell(\alpha) - \ell(\beta)} S^\beta$. The dual bases are related by $\mathtt{r}^*_{\mathtt{ru}} = \mathtt{r}^*_{\mathtt{r}\varepsilon} + \varepsilon^*_{\mathtt{r}\varepsilon}$ and $\mathtt{u}^*_{\mathtt{ru}} = \varepsilon^*_{\mathtt{r}\varepsilon}$.

Lemma 3 (Cut Lemma)**.** *For $m \geq 1$,*

$$D_m R_s = \sum_{i:\ P_i = \mathtt{r},\ P_{i+m} = \mathtt{u}} (-1)^{\#\mathtt{u}(P_{i+1} \cdots P_{i+m-1})}\ \mathrm{Ev}\big(P \text{ with the window } [i, i+m] \text{ replaced by } \varepsilon\big).$$

Proof. Let $\Omega_m : V^{\otimes(m+1)} \to V$ send $\mathtt{r}\, w\, \mathtt{u} \mapsto (-1)^{\#\mathtt{u}(w)} \varepsilon$ and every other $\{\mathtt{r}, \mathtt{u}\}$-word to 0. In tensor form $\Omega_m = \big(\mathtt{r}^* \otimes (\mathtt{r}^* - \mathtt{u}^*)^{\otimes(m-1)} \otimes \mathtt{u}^*\big)(\cdot)\, \varepsilon$ in the $\{\mathtt{r}, \mathtt{u}\}$-dual basis. By the change of dual bases this is $\big((\mathtt{r}^* + \varepsilon^*) \otimes (\mathtt{r}^*)^{\otimes(m-1)} \otimes \varepsilon^*\big)(\cdot)\, \varepsilon$ in the $\{\mathtt{r}, \varepsilon\}$-dual basis. So on $\{\mathtt{r}, \varepsilon\}$-words Ω_m sends $x\, \mathtt{r}^{m-1} \varepsilon \mapsto \varepsilon$ for $x \in \{\mathtt{r}, \varepsilon\}$, and all other words to 0.

Take a padded $\{\mathtt{r}, \varepsilon\}$-word, which Ev sends to S^β. A window $x\, \mathtt{r}^{m-1} \varepsilon$ ending at the right boundary ε of a block occurs exactly once for each block with at least m cells. If $x = \mathtt{r}$, the block has $\geq m + 1$ cells and the window removes m of them. If $x = \varepsilon$ (or x is the front padding letter, which Ev treats identically), the block has exactly m cells; it disappears and the neighbouring ε's merge. Blocks with $< m$ cells contribute nothing. This is exactly the derivation D_m on S^β, with $S_n \mapsto S_{n-m}$ and $S_0 = 1$. Hence $D_m \circ \mathrm{Ev} = \mathrm{Ev} \circ \sum_i \Omega_m^{(i)}$ on padded words. □

A *corner* is a factor $P_i P_{i+1} = \mathtt{ru}$. The Cut Lemma gives

$$D_1 = \sum_{\text{corners}} (\mathtt{ru} \to \varepsilon), \qquad D_2 = \sum_{\mathtt{rru} \text{ windows}} (\mathtt{rru} \to \varepsilon) - \sum_{\mathtt{ruu} \text{ windows}} (\mathtt{ruu} \to \varepsilon).$$

Lemma 4. $B = \sum_{\{c, c'\}} \big(\text{both corners } \mathtt{ru} \to \varepsilon\big) + \sum_{\mathtt{ruu} \text{ windows}} (\mathtt{ruu} \to \varepsilon)$*, the first sum over unordered pairs of distinct corners.*

Proof. Apply D_1 twice. Replacing a corner by $y \in \{\mathtt{r}, \mathtt{u}\}$ keeps all other corners (corners are disjoint factors) and creates at most one new corner: (P_{i-1}, y) with $P_{i-1} = \mathtt{r}$, $y = \mathtt{u}$, coming from an $\mathtt{rru}$ window; or (y, P_{i+2}) with $y = \mathtt{r}$, $P_{i+2} = \mathtt{u}$, coming from an $\mathtt{ruu}$ window. Each such window arises exactly once. Validity (first letter $\mathtt{r}$, last letter $\mathtt{u}$) depends only on the end letters, so it can be checked at the end. Hence $D_1^2 = 2 \sum_{\text{pairs}} + \sum_{\mathtt{rru}} (\to \varepsilon) + \sum_{\mathtt{ruu}} (\to \varepsilon)$. Subtract D_2 and halve. □

3. Proof of Theorem 1

Block coordinates. Write $P = \mathtt{r}^{a_1} \mathtt{u}^{b_1} \cdots \mathtt{r}^{a_m} \mathtt{u}^{b_m}$ with all $a_i, b_i \geq 1$; the i-th corner ends $\mathtt{r}^{a_i}$. Let X_i, Y_i be Ev of the $\mathtt{rru}$ window in block i (exists iff $a_i \geq 2$) and of the $\mathtt{ruu}$ window (iff $b_i \geq 2$), each replaced by ε, and let $C = \sum_{i<j} \mathrm{Ev}(\text{corners } i, j \to \varepsilon)$. Then

$$A = \sum_{a_i \geq 2} X_i - \sum_{b_i \geq 2} Y_i, \qquad B = C + \sum_{b_i \geq 2} Y_i.$$

Every $\mathrm{Ev}(\cdots \varepsilon \cdots)$ above is a sum of distinct ribbons with coefficient 1 (a 0/1-vector in the ribbon basis), and is nonzero since $L \geq 3$.

Top graded pieces. Let $U = \#\mathtt{u}(P)$. Every term of A and B has $\#\mathtt{u} \le U$, and the $\#\mathtt{u} = U$ components are

$$A_U = \sum_{i \in I_A} w(a - 2e_i), \qquad B_U = C_U = \sum_{i<j \in I_B} w(a - e_i - e_j),$$

where $w(a') = \mathtt{r}^{a'_1}\mathtt{u}^{b_1} \cdots \mathtt{r}^{a'_m}\mathtt{u}^{b_m}$, $I_A = \{i : a_i \ge 2,\ \neg(i = 1 \wedge a_1 = 2)\}$ and $I_B = \{i : \neg(i = 1 \wedge a_1 = 1)\}$ (the exclusions are windows at position 0, where $y = \mathtt{u}$ is invalid). For fixed b, $a' \mapsto w(a')$ is injective, so $\operatorname{supp} A_U \cap \operatorname{supp} C_U = \emptyset$. Mirror-symmetrically (counting $\#\mathtt{r}$, top value $V = \sum a_i$), A^V and C^V have disjoint supports and $B^V = -A^V + C^V$, with exclusion sets $I'_A = \{i : b_i \ge 2,\ \neg(i = m \wedge b_m = 2)\}$ and $I'_B = \{i : \neg(i = m \wedge b_m = 1)\}$.

Separation. If $A \ne 0$ and $C_U \ne 0$ or $C^V \ne 0$, then A, B are independent: from $\lambda A + \mu B = 0$, project to the top piece and use disjoint supports to get $\mu = 0$, then $\lambda = 0$. Now $C_U \ne 0$ unless $m = 1$ or $(m = 2, a_1 = 1)$, and $C^V \ne 0$ unless $m = 1$ or $(m = 2, b_2 = 1)$. Both fail only for hooks $P = \mathtt{r}^a\mathtt{u}^b$ and for the family (T) $P = \mathtt{r}\mathtt{u}^b\mathtt{r}^a\mathtt{u}$.

When $A = 0$ (Corollary 2). If $I_A \ne \emptyset$ then $A_U \ne 0$; if $I'_A \ne \emptyset$ then $A^V \ne 0$. Both are empty iff $a = (a_1, 1, \ldots, 1)$, $b = (1, \ldots, 1, b_m)$ with $a_1, b_m \le 2$: the four alternating words. For $(a_1, b_m) = (1, 1)$ there are no windows, so $A = 0$. For $(2, 2)$, $A = X_1 - Y_m$ and both are forced to the same word, so $A = 0$. For $(2, 1)$ and $(1, 2)$, $A = X_1$ resp. $-Y_m$ is nonzero (odd alternating). These two cases with $A = 0$ are exactly the even alternating s. In hooks and (T), $A \ne 0$ (below).

Generic case ($m \ge 2$, not (T)). If s is not even alternating, $A \ne 0$ and separation gives $r_2 = 2$. If s is even alternating, $A = 0$, and $B \ne 0$: $B = C + \sum Y_i$ is a sum of 0/1-vectors in the ribbon basis, so it has nonnegative coefficients and no cancellation occurs; and C is a nonempty sum (at least two corners, each pair with a valid choice). So $r_2 = 1$.

Hooks $P = \mathtt{r}^a\mathtt{u}^b$, $a + b = L + 2 \ge 5$. Here $C = 0$, $A = [a \ge 2]X - [b \ge 2]Y$, $B = [b \ge 2]Y$. If $a = 1$ ($R_s = \Lambda_d$): $A = -Y$, $B = Y \ne 0$, $r_2 = 1$. If $b = 1$ ($R_s = S_d$): $B = 0$, $A = X \ne 0$, $r_2 = 1$. If $a, b \ge 2$: $X \ni \mathtt{r}^{a-1}\mathtt{u}^{b-1}$ and $\mathtt{r}^{a-2}\mathtt{u}^b$ (if $a \ge 3$), $Y \ni \mathtt{r}^{a-1}\mathtt{u}^{b-1}$ and $\mathtt{r}^a\mathtt{u}^{b-2}$ (if $b \ge 3$), so $X \ne Y$ because $a = b = 2$ is excluded, and $r_2 = 2$.

Family (T) $P = \mathtt{r}\mathtt{u}^b\mathtt{r}^a\mathtt{u}$, $s = \mathtt{u}^b\mathtt{r}^a$. With $W_1 = \mathtt{r}\mathtt{u}^b\mathtt{r}^{a-2}\mathtt{u}$, $W_2 = \mathtt{r}\mathtt{u}^{b-2}\mathtt{r}^a\mathtt{u}$, $W_3 = \mathtt{r}\mathtt{u}^{b-1}\mathtt{r}^{a-1}\mathtt{u}$ (pairwise distinct), $A = [a \ge 2]W_1 - [b \ge 2]W_2$ and $B = W_3 + [b \ge 2]W_2$. Then $A \ne 0$ and $W_3 \notin \operatorname{supp} A$, so $r_2 = 2$; and s is neither constant nor alternating. □

4. Computed checks

All computed, exact arithmetic, against an independent S-basis NSym engine; none is used in the proof.

- Cut Lemma, $m \le 4$, $L \le 9$: 4036/4036 cases agree (`check2.py`).
- Formulas for A and B, all s with $2 \le L \le 9$: 0 disagreements (`eng.py`).
- $\operatorname{rank}\{A, B\}$ equals the Theorem's prediction, and $A = 0 \iff s$ even alternating, for all s with $L \le 14$ (`scan.py`, `check2.py`); this agrees with the table behind the Q75 note.
- Explicit A, B for hooks and (T), $a, b \le 11$ (`fam.py`).

5. Prior art

Owner search (2026-11-13): not found. No paper we read gives a formula for $M^{\perp}_{(m)}$, or a Ψ/Φ-type skewing, on the *ribbon* basis of NSym. Closest:

- Tewari, arXiv:1403.0607 (EJC 2016), Thm. 5.19: a Murnaghan–Nakayama rule for the *product* $\Psi_n \cdot s_\alpha$ on the noncommutative Schur basis s_α of Bessenrodt–Luoto–van Willigenburg, not for skewing ribbons.

- Lam–Lauve–Sottile, arXiv:0908.3714: skew ribbons $R_{\gamma/\beta} = F_\beta \rightharpoonup R_\gamma$ via coproduct coefficients; their Remark 10 says nonzero skew ribbons "do not correspond to skew ribbon shapes in a simple way". No closed form for $M^\perp_{(m)} R_\gamma$; this is the framework the lemma lives in.
- Jing–Liu, arXiv:2606.15138: a skew Murnaghan–Nakayama rule for Hopf dual pairs, including $(\mathrm{NSym}, \mathrm{QSym})$ (Thm. 5.1, Cor. 5.2), stated in the s_α basis with multiplication by Ψ_r. The authors note explicitly that the ribbons are a different basis. This is the paper closest to ours.
- Gelfand–Krob–Lascoux–Leclerc–Retakh–Thibon, hep-th/9407124: the product rule $R_I R_J = R_{IJ} + R_{I \triangleright J}$ used for ε, but no skewing operators on ribbons.

The *commutative image* is classical: under $\mathrm{NSym} \to \mathrm{Sym}$, $M^\perp_{(m)} \mapsto p^\perp_m$ and R_s maps to a ribbon skew Schur function, so the image of the Cut Lemma is an instance of the skew Murnaghan–Nakayama rule (Macdonald, *Symmetric Functions and Hall Polynomials*, I.3 Ex. 11). What appears to be new is the noncommutative lift in this signed-window form. The search was not exhaustive: forward citations of Tewari and of Lam–Lauve–Sottile were not checked, nor were the papers of Hivert or Novelli–Thibon searched in full. I would not be surprised if it is folklore.